The spectrum of occurrence or prevalence of disease in a defined population includes sporadic, endemic, epidemic and pandemic.

Sporadic refers to a disease that occurs infrequently and irregularly without a geographic focus. Examples of sporadic diseases include tetanus, rabies, and plague.

Endemic disease is an infectious disease which is restricted to a population in a given geographical region only and the constant rate of presence for years.

Epidemic refers to an increase in the number of cases of a disease in a particular geographical region within a short span of time when compared to the previous year infection rate. Influenza (common cold) is a good example of a common epidemic disease.

Pandemic refers to an epidemic that has spread over several countries or continents, usually affecting a large number of people i.e. a wide geographical region. AIDS is an example for pandemic since it is present in many countries.

Transmission

Transmission of an agent causing an infectious disease can be direct or indirect. The transfer of an infectious agent directly into the body is known as direct transmission. There are four types of **direct contact transmission**.

1. **Physical contact** between hosts (Influenza, Skin infections).
2. **Direct contact** with body fluids or tissues of an infected individual (HIV, HPV).
3. **Droplet contact** in which large infectious particles sprayed into the air from the respiratory tract of an infected individual (pneumonia, mumps, measles).
4. **Droplet nuclei contact** in which small infective dried droplet particles that are suspended in the air are taken in by a host, and are capable of traveling to the lung (TB, chickenpox).

Indirect transmission is the transfer of a pathogen by a vector or vehicle. Malaria is an example of a vector borne disease. Examples of diseases spread through vehicle-borne transmission are food-borne diseases and waterborne diseases eg: Cholera. Zoonosis occurs when diseases are transferred from animals to people. Zoonotic diseases include anthrax from sheep and plague from rodents.

Etiology

Etiology is the study of cause or origin of disease. The etiologic agent or causative agent is responsible for the cause of a disease. A pathogen or infectious agent is a biological agent that causes disease or illness to its host organism. Pathogenic organisms are of five major types - bacteria, virus, fungi, worms, and protozoa.

Diagnosis

Laboratory tests may identify organisms directly (e.g., visually, using a microscope, growing the organism in culture) or indirectly (e.g., identifying antibodies to the organism). They use a sample of blood, urine, sputum, stool, throat swab or other fluid or tissue from

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